

Notes: Hermalin & Wallace (JFE, 2001)

Firm Performance and Executive Compensation in the Savings and Loan Industry

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1 Big picture

- **Question.** How should researchers estimate the relation between CEO pay and firm performance when agency theory predicts that different firms choose different compensation contracts?
- **Core critique.** Standard pay-performance regressions impose the same performance coefficient across all firms, even though optimal incentives should vary with firm risk, size, capitalization, managerial ownership, and managerial ability.
- **Empirical setting.** The paper studies 104 publicly traded savings and loan institutions from 1988 to 1993.
- **Main method.** The paper uses a random-coefficients framework estimated by restricted maximum likelihood (REML), allowing each thrift to have its own pay-performance coefficients.
- **Main finding.** Once heterogeneity is modeled, the pay-performance relation is stronger and more informative than in standard fixed-effects OLS.
- **Economic takeaway.** The paper is about how firms design incentive systems, not merely whether pay rises with performance on average.

2 Data and measurement

2.1 Sample and sources

The data are an unbalanced panel of 104 publicly traded savings and loan institutions from 1988 through 1993, with 509 thrift-year observations in the baseline compensation regression. Sources include SNL Securities, Office of Thrift Supervision call reports, and CRSP.

Detailed interpretation. A single-industry setting reduces unrelated cross-industry differences, so evidence of heterogeneity within thrifts is especially informative.

2.2 Compensation and performance

CEO compensation includes salary, bonus, deferred compensation, and a binomial valuation of stock options granted. The dependent variable is log total compensation in the year after performance is measured. Performance is measured by ROA and annual percentage stock-price change.

Detailed interpretation. Including option values makes the compensation measure closer to total incentive compensation than salary-only measures.

2.3 Heterogeneity variables

The heterogeneity variables are stock-return standard deviation, production efficiency, tangible capital ratio, officer stock ownership, and assets.

Detailed interpretation. These variables are not just controls. They are theory-motivated determinants of incentive intensity.

3 Models and empirical specifications

3.1 Standard homogeneous model

$$y_j = x_j' \beta + z_j' c + e_j.$$

Detailed interpretation. This model assumes all firms have the same pay-performance coefficient.

3.2 Firm-specific coefficients

$$y_j = x_j' \beta_j + z_j' c + e_j, \quad \beta_j = z_j^{*'} \delta + \omega_j.$$

Detailed interpretation. Each firm can use a different compensation scheme, and observed heterogeneity variables explain part of that scheme.

3.3 Combined random-coefficients model

$$y_j = x_j' z_j^{*'} \delta + z_j' c + (e_j + x_j' \omega_j).$$

Detailed interpretation. REML is used because simple interacted OLS can be biased when performance is correlated with unobserved incentive intensity.

3.4 Contract-feature system

$$Pr(\text{feature } h \text{ in firm } j) = \Phi(z_j^{*'} \eta_h).$$

Detailed interpretation. The paper checks whether heterogeneity variables predict formal CEO compensation features, not just realized pay sensitivity.

4 Identification and empirical design

The paper identifies firm-specific incentive sensitivities and how they vary with theory-motivated firm characteristics. The setting is not a randomized treatment design; it is an agency-theory test. Identification comes from panel variation in compensation and performance, firm fixed effects, random coefficients, and cross-firm heterogeneity in size, risk, capitalization, ownership, and efficiency.

5 Section-by-section study guide

- **Introduction.** Explains why homogeneous pay-performance regressions are incomplete.
- **Theory and standard regression.** Shows how pooled OLS can distort typical incentives.
- **Random coefficients.** Develops the REML model and firm-specific incentive estimates.
- **Contract features.** Tests whether heterogeneity variables predict ex ante compensation-package structure.
- **Appendices.** Define data, REML estimation, and option valuation.

6 Tables and figures

6.1 Table 1: Fixed-effects OLS estimates of the compensation-performance relation

Read: Stock-price change is positive and significant; ROA is positive but insignificant. Firm fixed effects are strongly significant.

Detailed interpretation. Table 1 is the homogeneous benchmark that the paper later shows is incomplete.

Economic message. A pooled coefficient can hide firm-level incentive design.

Table 1
Fixed-effects OLS estimates of the compensation-performance relation. This regression employs 509 thrift-years. The dependent variable is the natural log of total compensation paid the CEO in the year following the year in which the independent variables were measured. Data are from SNL Securities for 104 publicly traded savings and loans. Standard errors are in parentheses. * denotes significance at the 10% level, ** denotes significance at the 5% level, and *** denotes significance at the 1% level.

Independent variables	Coefficient estimates
Assets	1.77×10^{-9} (2.04×10^{-9})
RAO (rate of return on assets)	1.44×10^{-3} (0.292)
Yearly percentage change in stock price	$2.31 \times 10^{-3}***$ (3.21×10^{-4})
R^2	0.82
Test for firm-fixed effects, $F_{106,402}$	16.86***

Table 2
Determinants of the relation between compensation and performance. The dependent variable is the natural logarithm of total CEO compensation in the year following the year in which the independent variables were measured. Data are from SNL Securities and CRSP for 104 publicly traded savings and loans. Method of estimation is REML. Standard errors are in parentheses. * denotes significance at the 10% level, ** denotes significance at the 5% level, and *** denotes significance at the 1% level.

Independent variable	Coefficient estimates
Intercept	12.1*** (2.02)
Standard deviation of stock return	2.35** (1.18)
Production efficiency	3.78×10^{-3} (0.097)
Tangible capital	-0.032** (0.014)
Share of stock owned by officers	-5.71×10^{-3} (0.007)
Assets	$7.01 \times 10^{-5}***$ (8.12×10^{-6})
ROA	0.024** (0.012)
Standard deviation of stock return $\times$ ROA	-0.014 (0.095)
Production efficiency $\times$ ROA	-0.003 (0.007)
Tangible capital $\times$ ROA	$-2.55 \times 10^{-3}***$ (0.001)
Share of stock owned by officers $\times$ ROA	5.06×10^{-4} (3.00×10^{-4})
Assets $\times$ ROA	-1.00×10^{-6} (8.11×10^{-7})
Percentage change in stock price	$3.55 \times 10^{-3}***$ (1.8×10^{-3})
Std. deviation of stock return $\times$ %age change in stock price	-0.018*** (0.008)
Production efficiency $\times$ percentage change in stock price	3.49×10^{-4} (8.0×10^{-4})
Tangible capital $\times$ percentage change in stock price	$1.55 \times 10^{-4}***$ (1.0×10^{-4})
Share of stock owned by officers $\times$ %age change in stock price	$1.58 \times 10^{-6}***$ (3.77×10^{-5})
Assets $\times$ percentage change in stock price	$2.0 \times 10^{-7}***$ (8.19×10^{-8})

6.2 Table 2: Determinants of the relation between compensation and performance

Read: Heterogeneity variables affect compensation levels and performance sensitivities. ROA and stock-price change both have positive baseline effects.

Detailed interpretation. Table 2 is the central evidence that incentive slopes vary systematically across firms.

Economic message. Contract heterogeneity is central to pay-performance estimation.

Table 2

Determinants of the relation between compensation and performance. The dependent variable is the natural logarithm of total CEO compensation in the year following the year in which the independent variables were measured. Data are from SNL Securities and CRSP for 104 publicly traded savings and loans. Method of estimation is REML. Standard errors are in parentheses. * denotes significance at the 10% level, ** denotes significance at the 5% level, and *** denotes significance at the 1% level.

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Table 2: Determinants of pay-performance relation

6.3 Figures 1 and 2: Firm-specific ROA and stock-price sensitivity estimates

Read: The random-coefficients estimates are more concentrated and less erratic than firm-by-firm OLS estimates.

Detailed interpretation. The REML approach recovers a clearer distribution of firm-level incentives.

Economic message. Firm-by-firm OLS is too noisy; model-based heterogeneity improves inference.

performance measures, change in annual stock price.

Another way to assess our results is to consider the mean impact of a one standard-deviation change in a performance measure on CEO pay. For ROA, this impact is \$8,066. For change in stock price the impact is \$52,791. Relatively to mean compensation, these represent 3% and 18% increases in

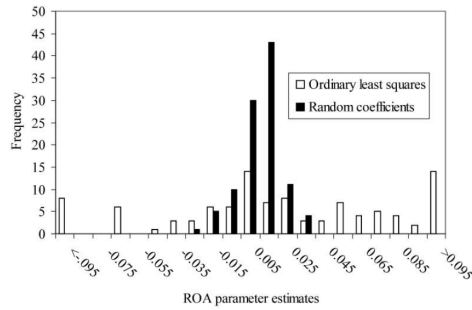


Fig. 1. Comparison of the firm-specific ROA parameter estimates from the random-coefficients specification and the ordinary least squares specifications.

(a) Figure 1: ROA parameter estimates

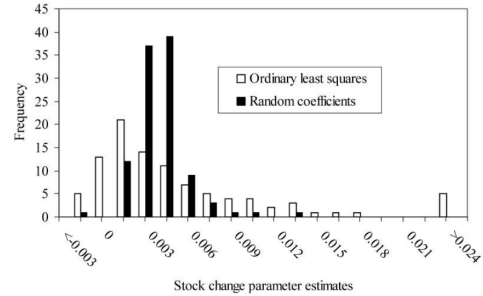


Fig. 2. Comparison of the firm-specific stock price change parameter estimates from the random-coefficients specification and the ordinary least squares specification.

(b) Figure 2: Stock-price-change parameter estimates

6.4 Figures 3 and 4: OLS versus random-coefficients trade-off

Read: OLS estimates show little pattern; random-coefficients estimates show a downward relationship between ROA and stock-price sensitivities.

Detailed interpretation. Firms trade off accounting and market performance measures.

Economic message. A weak average coefficient can mask strong but heterogeneous incentives.

Table 3
Nonlinear SUR estimates of the factors affecting CEO employment benefits. In each column the dependent variable is whether the employment contract contains that benefit. Data are from SNL Securities and CRSP for 104 publicly traded savings and loans. Method of estimation is nonlinear seemingly unrelated regression (SUR). Standard errors are in parentheses. * denotes significance at the 10% level, ** denotes significance at the 5% level, and *** denotes significance at the 1% level.

Independent variables	Employment contract	Golden parachute	Incentive plan	Restricted stock plan	Stock-option plan	Supplemental pension plan
Intercept	0.703 (0.426)	0.176 (0.315)	0.675** (0.304)	-0.823** (0.347)	-1.51* (0.766)	-1.19*** (0.344)
Assets	2.0×10^{-3} (0.0001)	3.0×10^{-3} (4.0 $\times 10^{-5}$)	0.0001* (0.0 $\times 10^{-5}$)	6.0×10^{-3} *** (0.0 $\times 10^{-5}$)	0.012*** (0.004)	2.0×10^{-3} *** (0.0 $\times 10^{-5}$)
Std. deviation of stock return	6.61** (3.36)	3.91* (2.29)	-1.37 (1.85)	0.62 (2.13)	-3.33 (2.63)	1.70 (2.00)
Production efficiency	0.769* (0.396)	0.139 (0.195)	-0.210 (0.176)	-0.149 (0.218)	0.699** (0.353)	0.196 (0.207)
Tangible capital ratio	0.024 (0.040)	0.038 (0.030)	-0.060** (0.027)	-0.041 (0.036)	0.062 (0.062)	-0.034 (0.034)
Share of stock owned by officers	-0.078*** (0.018)	-0.028*** (0.010)	-0.0001 (0.012)	-0.016 (0.018)	0.095* (0.056)	-0.021 (0.015)
Wald test	$\chi^2_1 = 53.29$ ***					

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Table 4
Summary of findings

Theoretical prediction	Empirical evidence
CEO pay will be positively related to performance measures.	104/104 thrifts positively weight at least one performance measure and 70/104 positively weight both performance measures.
There will be heterogeneity in the pay-for-performance relationship across thrifts.	Heterogeneity controls jointly significant in both REML (Table 2) and systems estimation (Table 3). Also observe dispersion in Figs. 1, 2, and 4.
Thrifts will make tradeoffs between the performance measures for the purposes of providing incentives.	Negative relationship between β s in Fig. 4.
Larger thrifts will use stronger incentives.	Strong evidence in Table 3. But no evidence in Table 2.
Noise in performance measure will affect strength of incentives. Theoretically, direction of effect indeterminant.	Evidence in Table 2 for effect. Apparent direction is that noise decreases strength of incentives.
Capitalization affects thrifts' strategies, which affects their incentive contracts.	Strong evidence in Table 2. Curiously, sign of effect is different for the two performance measures.
Managerial ability should affect incentives.	No evidence found, although proxy for ability could be overly crude.

6.5 Table 3: Nonlinear SUR estimates of the factors affecting CEO employment benefits

Read: Heterogeneity variables jointly predict six CEO contract features. Assets and efficiency are especially important for some features.

Detailed interpretation. Contract-feature evidence corroborates the random-coefficients results.

Economic message. Heterogeneity appears in actual compensation package design.

6.6 Table 4: Summary of findings

Read: The table maps theoretical predictions to empirical evidence.

Detailed interpretation. It summarizes which agency-theory predictions are supported, weak, or anomalous.

Economic message. The results support a richer, heterogeneous-contract view of incentive pay.

6.7 Tables 5 and 6: Summary statistics and option values

Read: Table 5 describes the sample and contract features; Table 6 reports annual option values.

Detailed interpretation. Option grants are economically meaningful and should be included in compensation.

Economic message. The dependent variable includes important incentive-pay components.

Table 5
Summary statistics for a sample of 104 thrift institutions, 1988–1993

Variables	Mean	Standard deviation
<i>Panel A: Data from SNL Securities</i>		
Compensation of CEO (1988 dollars)	\$287,218	\$271,713
Total assets (1988 dollars)	\$29,826,807	\$69,260,715
Employment contracts	84.8%	35.8%
Golden parachutes	72.5%	44.6%
Incentive plans	56.7%	49.7%
Restricted stock plans	15.6%	36.3%
Stock option plans	94.0%	23.0%
Supplemental pension plan	22.8%	42.0%
Tangible capital (equity ÷ assets)	0.062	0.038
Share of stock owned by officers	4.63%	6.17%
Rate of return on assets (ROA)	0.36%	0.83%
Annual change in stock price	5.20%	34.7%

(a) Table 5: Summary statistics

Table 6
Estimated stock option values in 1988 dollars for CEOs in our sample of thrift institutions

Year	Average per CEO	Standard deviation
1989	\$48,040	\$150,626
1990	\$42,426	\$103,057
1991	\$15,472	\$40,071
1992	\$13,104	\$38,824
1993	\$35,762	\$70,137

(b) Table 6: Option values

7 Short summary

- The paper argues that standard pay-performance regressions are incomplete because they impose a common incentive slope across firms.
- A random-coefficients model reveals stronger and more systematic pay-performance incentives.
- Firms trade off ROA-based and stock-price-based performance measures.
- Heterogeneity variables also predict formal contract features.
- The paper’s main lesson is that empirical agency tests should model contract heterogeneity directly.

8 Expanded conclusion

8.1 Core conclusion

Hermalin and Wallace show that executive pay-performance sensitivity cannot be understood well if all firms are assumed to use the same compensation scheme.

8.2 Economic intuition

Agency theory predicts heterogeneous incentive contracts because firms differ in risk, size, information, capital structure, managerial ownership, and managerial ability.

8.3 Methodological contribution

The paper uses random coefficients and REML to estimate firm-specific incentive sensitivities while explaining them with heterogeneity variables.

8.4 Governance interpretation

The contract-feature analysis shows that heterogeneity affects not only realized pay but also the architecture of CEO compensation packages.

8.5 Limitations

The sample is small, industry-specific, and the ability and ownership proxies are imperfect. Many heterogeneity variables may also be endogenous in the long run.

8.6 Takeaway

The question is not only whether pay responds to performance, but how different firms choose different performance measures and incentive strengths.